

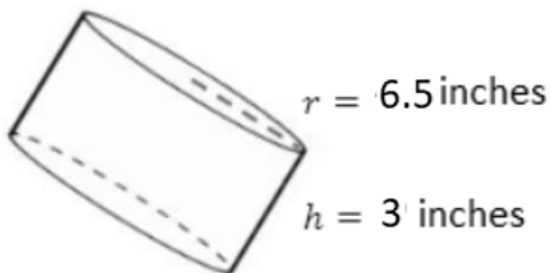
Grade 7
Unit 12
Lesson 5 Practice

Name _____
Date _____
Period _____

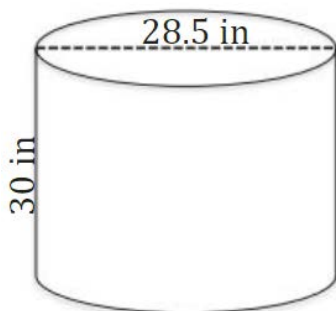
The dimensions of four cylinders are given in the column on the left. Match each set of dimensions with the corresponding volume of the cylinder.

- | | |
|--|--------------------------------|
| 1. _____ $r = 7$ units; $h = 4$ units | A. 12π units ³ |
| 2. _____ $r = 3$ units; $h = 7$ units | B. 63π units ³ |
| 3. _____ $d = 4$ units; $h = 3$ units | C. 144π units ³ |
| 4. _____ $d = 12$ units; $h = 4$ units | D. 196π units ³ |

5. Calculate the volume of the cylinder. Use 3.14 for π and round to the nearest thousandths place.



6. Calculate the volume of the cylinder. Leave the volume in terms of π .



Gunnar has a watering bowl that is shaped like a right circular cylinder. The bowl is 10 inches (in.) in diameter and has a height of 6.5 in.

7. Draw a picture of the bowl and label the dimensions.
8. How many cubic inches of water can the bowl hold? Use $\frac{22}{7}$ for π and round your answer to the nearest tenths place.

Sammy calculated the volume of a cylinder with a diameter of 9 feet (ft.) and height of 20 ft. His calculations are shown.

$$\begin{array}{r}(\pi \cdot 9^2)(20) \\ (81\pi)(20) \\ 1620\pi\end{array}$$

9. Did Sammy calculate the volume correctly? If not, describe the mistake he made.
10. Find the volume and leave your answer in terms of π .